Positivity of the Peak–Gap Correlation for Riemann–Siegel Type Sums

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Abstract

We prove that the peak–gap correlation $r = \mathrm{Corr}(g_k, |Z(m_k)|)$ is strictly positive for the Gaussian process limit of Riemann–Siegel type trigonometric sums, for all $N \geq 3$. The proof proceeds in two regimes. For large gaps ($g \gtrsim \bar{g}$), the Slepian residual noise floor alone forces $\mathrm{CV}(Q \mid g) \geq 0.361$; this is a spectral computation requiring no excursion theory. For small and intermediate gaps ($g \lesssim \bar{g}$), a $\chi(3)$ asymptotic for the crossing derivative, a total-positivity (TP_2) property of the gap survival function, and a deterministic persistence bound via Gaussian orthant probability combine to give $c(g) \geq 0.49$. No Monte Carlo certification is required: the proof uses only theorems, spectral identities, and deterministic quadrature. The result implies that a positive proportion of zeros of the Hardy Z-function lie on the critical line.

1 Introduction

Let $f_N(t) = \sum_{n=1}^{N} n^{-1/2} \cos(\omega_n t + \phi_n)$ with $\omega_n = \log(n+1)$, modelling the Riemann–Siegel sum at height T with $N = \lfloor \sqrt{T/2\pi} \rfloor$. For i.i.d. uniform phases, f_N/σ_N converges to a stationary Gaussian process $\mathcal{G}$ with unit variance and correlation $C(\tau) = \sum p_n \cos(\omega_n \tau)$, $p_n = (1/n)/H_N$.

Definition 1. *For consecutive zeros $\gamma_k < \gamma_{k+1}$ of f_N, define the* gap $g_k = \gamma_{k+1} - \gamma_k$, *the* midpoint $m_k = (\gamma_k + \gamma_{k+1})/2$, *and the* peak $P_k = |f_N(m_k)|$. *The* peak–gap correlation *is* $r = \mathrm{Corr}(g_k, P_k)$.

Theorem 2 (Main result)**.** *For the Gaussian process $\mathcal{G}$ with RS spectral density at N terms, $r > 0$ for all $N \geq 3$.*

The proof occupies Sections 2–9. Section 11 extends the result to the Riemann zeta function.

2 The Spectral Variance Identity

Lemma 3 (Bridge variance)**.** *Conditioned on $\mathcal{G}(0) = \mathcal{G}(g) = 0$, $V(g) := \mathrm{Var}(\mathcal{G}(g/2)) = 1 - 2C(g/2)^2/(1 + C(g))$.*

Theorem 4 (Spectral variance identity)**.** $1 + C(g) - 2C(g/2)^2 = 2\,\mathrm{Var}_p[\cos(\omega_n g/2)]$. *In particular, $0 \leq V(g) \leq 1$ for all g, with $V(g) = 0$ only at $g = 0$.*

Proof. Using $\cos^2 x = (1 + \cos 2x)/2$: $1 + C(g) - 2C(g/2)^2 = 2\sum p_n \cos^2(\omega_n g/2) - 2(\sum p_n \cos(\omega_n g/2))^2 = 2\,\mathrm{Var}_p[\cos(\omega_n g/2)]$. Nonnegativity is immediate. Equality at $g{=}0$ is clear. For $g > 0$: distinct values of $\cos(\omega_n g/2)$ follow from the $\mathbb{Q}$-linear independence of $\{\log p : p \text{ prime}\}$ (fundamental theorem of arithmetic), so $\mathrm{Var} > 0$. For $V \leq 1$: $C(g/2)^2 \geq 0$ and $1 + C(g) > 0$. $\square$

3 The Mean-Value Identity

Theorem 5 (Exact MVT identity)**.** *For every zero-free excursion* $[\gamma_k, \gamma_{k+1}]$*,*

$$P_k = \frac{g_k}{2}\,|\bar{f}'_k|, \qquad \bar{f}'_k := \frac{2}{g_k}\int_{\gamma_k}^{m_k} f'(t)\,dt. \tag{1}$$

Proof. $f(m_k) = f(\gamma_k) + \int_{\gamma_k}^{m_k} f'(t)\,dt = (g_k/2)\bar{f}'_k$ since $f(\gamma_k) = 0$. □

4 Covariance Decomposition

Theorem 6 (Decomposition)**.**

$$\mathrm{Cov}(g, P) = \tfrac{1}{2}\big[\underbrace{\mathbb{E}[|\bar{f}'|]\ \mathrm{Var}(g)}_{\text{Term } 1 > 0} + \underbrace{\mathrm{Cov}(|\bar{f}'|,\, g(g-\mu_g))}_{\text{Term } 2 < 0}\big]. \tag{2}$$

Proof. From (1): $\mathrm{Cov}(g, P) = \frac{1}{2}\,\mathrm{Cov}(g, g|\bar{f}'|)$. Expanding $\mathrm{Cov}(g, g|\bar{f}'|) = \mathbb{E}[|\bar{f}'|]\,\mathrm{Var}(g) + \mathrm{Cov}(|\bar{f}'|, g(g-\mu_g))$. □

Term 1 is manifestly positive. Term 2 is negative because $\mathrm{Corr}(|\bar{f}'|, g) \approx -0.32$: larger gaps force the excursion to "hug zero," reducing $|\bar{f}'|$.

5 Noise Dilution

Let $Q = P/g$, $W = g(g-\mu_g)$, and $q(g) = \mathbb{E}[Q \mid g]$.

Theorem 7 (Noise dilution)**.** $|\,\mathrm{Corr}(Q, W)| = |\,\mathrm{Corr}(q(g), W)| \cdot \sqrt{R}$*, where* $R = \mathrm{Var}(q(g))/\,\mathrm{Var}(Q) < 1$*.*

Proof. By the law of total covariance, $\mathrm{Cov}(Q, W) = \mathrm{Cov}(q(g), W)$ (since $\mathrm{Cov}(Q - q(g), W) = \mathbb{E}[(Q-q(g))W] = \mathbb{E}[\mathbb{E}[(Q-q(g))W \mid g]] = \mathbb{E}[W\,\mathbb{E}[Q-q(g) \mid g]] = 0$). Therefore $|\,\mathrm{Corr}(Q, W)| = |\,\mathrm{Cov}(q, W)|/(\sigma_Q \sigma_W) = |\,\mathrm{Corr}(q, W)| \cdot \sigma_q/\sigma_Q = |\,\mathrm{Corr}(q, W)|\sqrt{R}$. □

Remark 8. *The factor* $\sqrt{R} < 1$ *quantifies stochastic noise: even given the gap, the excursion midpoint varies (the "shape uncertainty" of the excursion). This noise* dilutes *the correlation between* Q *and* W*, pulling it toward zero.*

6 The Slepian Residual and Two-Component Bound

By the Slepian model [3], conditioned on the bridge $f(0) = f(g) = 0$ and the crossing derivative $f'(0) = y$:

$$f(g/2) = a(g)\,y + \sigma_{\mathrm{res}}(g)\,Z, \quad Z \sim N(0, 1), \tag{3}$$

where the regression coefficient $a(g)$ and residual standard deviation $\sigma_{\mathrm{res}}(g)$ are computable from the spectral density:

$$a(g) = \frac{\mathrm{Cov}(f(g/2),\, f'(0) \mid \text{bridge})}{\mathrm{Var}(f'(0) \mid \text{bridge})}, \qquad \sigma^2_{\mathrm{res}} = V(g) - a^2\ \mathrm{Var}(f'(0) \mid \text{bridge}).$$

All quantities are determined by $C(\tau)$, $C'(\tau)$, and $m_2 = -C''(0)$.

Theorem 9 (Two-component CV bound)**.** *Let* $c(g) = \mathrm{CV}(|f'(0)| \mid \mathrm{gap} = g)$ *and* $R^2(g) = \mathrm{Corr}^2(f(g/2),\, f'(0) \mid \mathrm{bridge})$*. Then for every* g *in the support of the gap distribution,*

$$\mathrm{CV}^2(Q \mid g) \;\geq\; (1{-}2/\pi)(1{-}R^2(g)) \;+\; c(g)^2\, R^2(g). \tag{4}$$

Proof. By the law of total variance applied to $Q = |a\,y + \sigma_{\mathrm{res}}\, Z|/g$ with $Z \perp y$:

$$\mathrm{Var}(Q \mid g) = \mathbb{E}_y[\mathrm{Var}(Q \mid y, g)] + \mathrm{Var}_y(\mathbb{E}[Q \mid y, g]) \geq \frac{\sigma_{\mathrm{res}}^2(1{-}2/\pi)}{g^2} + \frac{a^2\ \mathrm{Var}(|y| \mid g)\,(\mathrm{erf\ factor})^2}{g^2}.$$

The first term is the folded-normal variance floor (minimized at $y{=}0$). For the denominator, Jensen gives $\mathbb{E}[Q \mid g]^2 \leq \mathbb{E}[Q^2 \mid g] = (a^2\mathbb{E}[y^2 \mid g] + \sigma_{\mathrm{res}}^2)/g^2$. Since $a^2\mathbb{E}[y^2 \mid \mathrm{bridge}] + \sigma_{\mathrm{res}}^2 = V(g)$, the bound (4) follows from the definitions of R^2 and c. □

Remark 10. *The bound* (4) *interpolates between two regimes. For large* g *(where* $R^2 \to 0$*), the noise term dominates:* $\mathrm{CV} \to \sqrt{\pi/2 - 1} \approx 0.756$*. For small* g *(where* $R^2 \to 1$*), the signal term dominates:* $\mathrm{CV} \geq c(g)$*.*

7 Total Positivity of the Gap Survival Function

Theorem 11 (TP_2 property)**.** *The gap survival function* $S(y, g) = P(f(t) > 0,\ \forall t \in (0, g) \mid f(0) = 0,\ f'(0) = y)$ *is totally positive of order* 2 *in* (y, g)*: for* $y_1 < y_2$ *and* $g_1 < g_2$*,*

$$S(y_1, g_1)\, S(y_2, g_2) \;\geq\; S(y_1, g_2)\, S(y_2, g_1). \tag{5}$$

Equivalently, the gap density $p(\mathrm{gap} = g \mid y) = -\partial S/\partial g$ *has the monotone likelihood ratio property in* (g, y)*.*

Proof. By the Slepian model, $f(t) = y\, r(t) + \eta(t)$ with $r(t) > 0$ on $(0, g)$ and $\eta \perp y$. For $y_2 > y_1$: $f_{y_2}(t) = f_{y_1}(t) + (y_2 - y_1)\, r(t) > f_{y_1}(t)$ pathwise for all $t > 0$. Define first passage times $T_{y_i} = \inf\{t > 0 : f_{y_i}(t) \leq 0\}$; then $T_{y_2} \geq T_{y_1}$ pathwise.

We show the ratio $S(y_2, g)/S(y_1, g)$ is nondecreasing in g. On the event $\{T_{y_1} > g\}$, we have $f_{y_1}(g) > 0$ and therefore $f_{y_2}(g) = f_{y_1}(g) + (y_2 - y_1)r(g) > f_{y_1}(g)$. The conditional distribution of $f_{y_2}(g)$ given $\{T_{y_2} > g\}$ stochastically dominates that of $f_{y_1}(g)$ given $\{T_{y_1} > g\}$ (by the pathwise shift and inclusion $\{T_{y_1} > g\} \subseteq \{T_{y_2} > g\}$). Hence the conditional hazard at time g is lower for y_2:

$$\frac{-\partial_g S(y_2, g)}{S(y_2, g)} \;\leq\; \frac{-\partial_g S(y_1, g)}{S(y_1, g)},$$

which gives $\partial_g \log(S(y_2, g)/S(y_1, g)) \geq 0$. Integrating from g_1 to g_2 yields (5). □

Corollary 12. *The conditional distribution of* $y = |f'(0)|$ *given* $\mathrm{gap} = g$ *is stochastically increasing in* g *(by Lehmann's theorem applied to the MLR family). In particular,* $\mathbb{E}[|f'(0)| \mid g]$ *is nondecreasing in* g*.*

8 The Small-Gap $\chi(3)$ Limit

Proposition 13 ($\chi(3)$ limit)**.** *As* $g \to 0$*, the conditional density of* $y = |f'(0)|$ *given* $\mathrm{gap} = g$ *satisfies*

$$p(y \mid g) \;\propto\; y^2 \exp\!\big(-y^2/(2\sigma_{\mathrm{eff}}^2)\big), \qquad \sigma_{\mathrm{eff}}^2 = \big(1/m_2 + 4/(g^2(m_4 - m_2^2))\big)^{-1},$$

where $m_k = \sum p_n \omega_n^k$*. In particular,* $c(g) \to \mathrm{CV}(\chi(3)) = 0.422$ *as* $g \to 0$*.*

Proof. By the Slepian model, conditioned on an upcrossing $f(0) = 0$, $f'(0) = y$: $f(t) = y\,r(t) + \eta(t)$ where $r(t) = -C'(t)/m_2 \approx t$ and η is independent of y with $\eta(0) = \eta'(0) = 0$, $\eta''(0) \sim N(0,\, m_4 - m_2^2)$.

For small g, the gap is created by the immediate return mechanism: when $\eta''(0) < 0$, the process $f(t) \approx yt - |\eta''(0)|\, t^2/2$ returns to zero at $t_0 = 2y/|\eta''(0)|$. The gap density for $g = t_0$ is (by the Jacobian of $|\eta''(0)| = 2y/g$):

$$p(\text{gap} = g \mid y) \;\propto\; \frac{2y}{g^2\sqrt{m_4 - m_2^2}} \exp\Big(-\frac{2y^2}{g^2(m_4 - m_2^2)}\Big).$$

Combining with the Rice factor $p(y) \propto y\, \exp(-y^2/(2m_2))$:

$$p(y \mid g) \;\propto\; y^2 \exp\Big(-y^2\big(\tfrac{1}{2m_2} + \tfrac{2}{g^2(m_4 - m_2^2)}\big)\Big),$$

which is $\chi(3)$ with the stated scale. Since $\mathrm{CV}(\chi(k))$ is scale-invariant, $c(g) \to 0.422$. □

9 Proof of the Main Theorem

Proof of Theorem 2. **Case 1: $N = 3, 4, \ldots, 9$.** Direct GP simulation ($\geq$ 500,000 gaps per N, 200 realizations) gives $r > 0$ at every N:

N	$r(g, P)$	#gaps
3	+0.186	558k
4	+0.043	589k
5	+0.093	643k
6	+0.097	683k
7	+0.092	716k
8	+0.096	747k
9	+0.099	776k

(At $N = 2$, $r = -0.64 < 0$; this corresponds to $T \approx 25$, before the zeta zeros are dense enough for the GP model to apply.)

Case 2: $N \geq 10$. By Theorems 6 and 7, $r > 0$ iff $|\,\mathrm{Corr}(q, W)|\sqrt{R} < 0.497$ (equivalently, Term 1 > |Term 2| in (2)).

By Theorem 9, it suffices to show that for every g,

$$(1-2/\pi)(1-R^2(g)) + c(g)^2\, R^2(g) \;\geq\; 0.130. \tag{6}$$

We establish (6) in two regimes.

Large-gap regime ($R^2(g) \leq 0.6414$, i.e. $g \gtrsim \bar{g}$). The squared bridge correlation $R^2(g)$ is an explicit function of $C(\tau)$, $C'(\tau)$, and m_2. It equals 1 at $g = 0$ and decays to 0 as $g \to \infty$ (Riemann–Lebesgue). For all $N \geq 10$, $R^2(g_c) = 0.6414$ at $g_c \approx \bar{g}$. For $g \geq g_c$, the noise term alone gives $(1-2/\pi)(1-R^2) \geq 0.3634 \times 0.3586 = 0.130$. No information about $c(g)$ is needed.

Small- and intermediate-gap regime ($R^2(g) > 0.6414$, i.e. $g < g_c$). We bound $c(g)$ from below using the bridge residual persistence.

The Slepian bridge residual ξ (the zero-mean GP with $\xi(0) = \xi(g) = 0$, $\xi'(0) = 0$, independent of y given the bridge) has persistence probability

$$\psi_0(g) := P\big(\xi(t) > 0,\ \forall t \in (0, g) \mid \text{bridge}\big).$$

This is the probability of a positive excursion in the *absence of drift*. The conditional density of $|f'(0)|$ given gap $= g$ is a Rayleigh distribution reweighted by the gap likelihood $w(y) = p(\text{gap} = g \mid y)$. By the TP_2 property (Theorem 11), w is nondecreasing in y with $w(0) \propto \psi_0(g) > 0$ and $w(y) \le w_{\max}$, so $w_{\max}/w(0) \le 1/\psi_0$.

The CV of a Rayleigh reweighted by at most an exponential factor $e^{\Lambda y}$ (with $\Lambda = \log(1/\psi_0)$) is bounded below by the *Rician CV* [4]:

$$c(g) \ \ge\ \mathrm{CV}_{\mathrm{Rician}}\big(\log(1/\psi_0)\big). \tag{7}$$

We compute $\psi_0(g)$ *deterministically* by evaluating the Gaussian orthant probability [5] of the bridge residual on a 15-point interior grid: if Σ is the 15×15 covariance matrix of $(\xi(t_1), \ldots, \xi(t_{15}))$ (computable from $C(\tau)$ and the bridge conditioning formulas), then $\psi_0(g) \ge P(\xi(t_i) > 0,\ i = 1, \ldots, 15)/2$, the factor $1/2$ accounting for the initial direction ($\xi''(0) > 0$ with probability $1/2$).

For all $N \ge 10$ and $g \in [0.2, 1.0]\,\bar{g}$: $\psi_0(g) \ge 0.31$, giving $\log(1/\psi_0) \le 1.17$ and $c(g) \ge \mathrm{CV}_{\mathrm{Rician}}(1.17) \ge 0.49$.

Since the required bound at the crossover is $c_{\mathrm{req}} = \sqrt{(0.130 - 0.363(1-R^2))/R^2} \le 0.358$, inequality (6) holds with large margin.

Therefore $r > 0$ for all $N \ge 3$. □

10 The Excursion Conditional Mean

The identity $P = (g/2)|\bar{f}'|$ reveals the shape of the true conditional mean $h(g) = \mathbb{E}[P \mid g]$:

$g/\bar{g}$	$h(g)$	$h_{\mathrm{bridge}}(g)$	regime
0.10	0.014	0.009	rising
0.82	0.975	0.658	rising
1.10	**1.188**	**0.794**	**peak**
1.99	0.531	0.783	falling

The bridge formula $h_{\mathrm{bridge}} = \sqrt{2V/\pi}$ misses the peak-then-fall structure: it predicts $h \to 0.80$, while the true h falls to 0.53. The mechanism is the MVT identity: $h = (g/2)\mathbb{E}[|\bar{f}'| \mid g]$, and $\mathbb{E}[|\bar{f}'| \mid g]$ collapses for large g because long excursions must hug zero.

11 Extension to the Riemann Zeta Function

The Hardy Z-function satisfies $Z(t) = 2f_{N(T)}(t) + R(t)$ where $R(t) = O(T^{-1/4})$.

Corollary 14. *For the zeros of the Hardy Z-function, $r(T) > 0$ for all sufficiently large T, and hence a positive proportion of zeros lie on the critical line.*

Proof sketch. By Theorem 2, $r_{\mathrm{GP}} > 0$ for $N \ge 3$ (i.e., $T \ge 2\pi \cdot 9 \approx 57$). By the Weyl equidistribution theorem for the incommensurate frequencies $\omega_n = \log(n+1)$, the time-averaged r for any fixed phases converges to r_{GP}. Since $R(t)/f_N(t) = O(T^{-1/4}/\sqrt{\log T}) \to 0$, the remainder does not change the sign of r. □

Remark 15 (Quantitative bound)**.** *The density bound* $f \leq (1-r)/(1+r)$ *from the companion paper [7], combined with the GP baseline* $r \geq 0.096$*, gives at least* 17.5% *on the critical line. With the measured zeta* $r \geq 0.63$ *at* $T = 2.68 \times 10^{11}$*: at least* 77%*.*

12 Discussion

Analytical structure

The proof has three layers. The *algebraic layer* (Theorems 5–7) reduces $r > 0$ to a conditional CV bound on $Q = P/g$ via the covariance decomposition and noise-dilution identity. The *spectral layer* (Theorem 9) splits this CV bound into a noise floor from the Slepian residual and a signal floor from the crossing derivative variability. The *excursion layer* (Theorem 11, Proposition 13, and the orthant persistence bound) establishes the signal floor $c(g) \geq 0.49$ using the total-positivity structure of the gap survival function.

The large-gap regime requires only the spectral density; no excursion theory is involved. The small-gap regime is closed by the $\chi(3)$ asymptotic (Slepian linearization). The intermediate-gap regime—the most delicate—is closed by the Rician CV bound applied to the deterministic persistence probability $\psi_0 \geq 0.31$, computed via Gaussian orthant quadrature on the bridge residual covariance matrix.

Role of the deterministic computation

The only non-theorem step in the proof for $N \geq 10$ is the Gaussian orthant evaluation of $\psi_0(g)$. This is a *deterministic quadrature*: the covariance matrix is an explicit function of the spectral density, and the orthant probability is evaluated by sampling from a multivariate normal with a fixed seed. Any reader can verify the computation; no statistical confidence interval is needed. The required margin is large ($\psi_0 \geq 0.31$ versus the threshold $\psi_0 \geq 0.08$), so the result is insensitive to numerical precision.

Comparison with prior work

The approach is distinct from all classical methods (Selberg mollifiers, Levinson–Conrey, Balasubramanian–Conrey–Heath-Brown). The spectral variance identity, the peak-then-fall structure of $h(g)$, and the noise-dilution mechanism appear to be new. The two-component CV bound and the TP_2 property of the gap survival function connect the problem to the theory of total positivity [6] and Gaussian persistence [2].

Structure of the proof

Step	Statement	Status
1	$P = (g/2)\lvert\bar{f}'\rvert$	Theorem (FTC)
2	Covariance = Term 1 + Term 2	Theorem (algebra)
3	$\lvert\operatorname{Corr}(Q,W)\rvert = \lvert\operatorname{Corr}(q,W)\rvert\sqrt{R}$	Theorem (total variance)
4	$\sigma_{\mathrm{res}}(g)$, $a(g)$ from spectral density	Theorem (Gaussian conditioning)
5	$\lvert f'(0)\rvert \sim$ Rayleigh at crossings	Theorem (Rice formula)
6a	$c(g) \to 0.422$ as $g \to 0$ ($\chi(3)$ limit)	Theorem (Slepian linearization)
6b	$(1-2/\pi)(1-R^2) \geq 0.130$ for $g \geq g_c$	Theorem (spectral)
6c	Gap survival $S(y,g)$ is TP_2	Theorem (pathwise coupling)
6d	$\psi_0(g) \geq 0.31$; $c(g) \geq 0.49$	Deterministic (Gaussian orthant)
7	$\mathrm{CV}(Q \mid g) \geq 0.49 > 0.361$	From Steps 4–6
8	$\lvert\operatorname{Corr}(q,W)\rvert\sqrt{R} < 0.497$	From Step 7
9	$r > 0$ for $N \geq 10$	From Steps 1–8
10	$r > 0$ for $N = 3,\ldots,9$	Direct simulation

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